ASTROVET

Technical Deep-Dive & Viva Preparation

A code-grounded review of the TESS light-curve vetting workflow, its models, evidence, evaluation, limitations, and next validation steps.

Implementation status and metrics are labeled by evidence level. Historical results are not presented as freshly reproduced.

Evidence boundary

- Verified means inspected in this checkout/source or local artifact.
- Recorded means an earlier project report value, not freshly reproduced.
- Not verified means referenced data/model or execution was unavailable.
- Checkout is `C:\Users\kalap\Documents\MountE\Final Year Projet\ExoMiner-main\astrovet`. The historical `D:\Final Year Projet\ASTROVET_data` path is unavailable here. Raw CSV and documented full fusion bundle are absent from this checkout.

1. Project understanding

Simple viva answer: ASTROVET reads a TESS light-curve FITS, audits and cleans measurements, searches for recurring transit-like dips, computes physical consistency checks, optionally encodes the curve with self-supervised learning, and presents evidence for human review. A separate supervised fusion model was trained on a selected TOI disposition cohort. NASA ExoMiner/ExoMiner++ is an optional external comparison, not rewritten. Groq optionally rephrases a compact summary.

Technical answer: Input is time, flux, uncertainty and quality values from SPOC/QLP FITS plus optional candidate period, epoch, duration, depth and target metadata. Workflow: FITS validation; cadence filtering; segment-aware normalization/detrending; BLS/TLS; phase folding; approximate transit fit, odd/even, secondary, plausibility checks; optional 16-D SSL representation; optional baseline; local score/report. Missing branches should remain unavailable. Output is triage evidence, not astrophysical confirmation.

Problem/users: TESS produces many transit-like events. Dips may be eclipsing binaries, variability, systematics or noise. Intended users are researchers, students and human vetters. The project seeks consistent evidence presentation; reduced human workload has not been measured.

Contribution/novelty: Evidence supports an integrated, auditable prototype built from established methods. It does not support claims of a new BLS/TLS algorithm or proven superiority to ExoMiner++. Contribution: FITS workflow, parallel searches, diagnostics, SSL representation, supervised fusion prototype, external baseline integration, reports and UI.

Boundaries: It does not control telescopes, conduct follow-up or formally validate planets. Internet is needed for MAST/TESSCut/Groq; Docker is for optional live ExoMiner. Pixel cutouts are context, not time-series model input.

Implemented, partial, unverified

Capability	Status/evidence
FITS upload validation/audit	Implemented: `src/astrovet/lightcurves/validator.py`, `src/astrovet/ui/server.py`; representative rejection cases in `docs/validation_results.md`.
SPOC/QLP reader, quality filtering, normalization	Implemented: `lightcurves/reader.py`, `quality.py`, `normalization.py`; archive-wide support unproven.
BLS/TLS and physical screens	Implemented: `views/periodogram.py`, `physics/*.py`; screens are heuristics.
Self-supervised encoder	Implemented, checkpoints exist: `ssl/neural_encoder.py`; three small real-FITS NPZ checkpoints present, not the documented 11,511-file checkpoint.
Catalog E-vs-B/J classifier	Implemented; `artifacts/models/tce_e_vs_bj_classifier.joblib` exists; not planet identification.
TOI prototype/fusion head	Training code exists; historic metrics documented. Matching full fusion bundle absent here.
NASA baseline	External adapter/catalog and container workflow exist; local catalog absent. Prior report describes OOM.
Dashboard/report/pixel view	Implemented: `ui/server.py`, `ui/static/index.html`, `explain/report.py`, `pixel_preview.py`; parity evidence covers two files only.
Groq explanation	Optional `explain/groq_explainer.py`; live response not verified; cannot change result.
Broad scientific validation	Incomplete; blind cohort and injection grid pending in `docs/validation_results.md`.

2. Algorithms and mathematical details

Logistic regression (catalog model)

`src/astrovet/models/training.py` uses `sklearn.linear_model.LogisticRegression`. For features `x`, weights `w`, intercept `b`: `z=w■x+b`, `p=σ(z)=1/(1+exp(-z))`. The library minimizes regularized, class-weighted binary log loss (normalization is library-defined): `Σ■ s■[-y■log(p■)-(1-y■)log(1-p■)] + λ||w||²■`.

Learned parameters are coefficients/intercept. Hyperparameters include `C`, solver, iterations, class weight, seed. `C` is inverse regularization (higher C means weaker penalty). Catalog run uses `lbfgs`, `max_iter=2000`, seed 42, balanced class weights; validation tests C `[.001,.01,.1,.5,1,5,10]` by PR-AUC, recorded winner 10. The optimizer is sklearn's L-BFGS.

`TabularFeaturePreprocessor` creates 12 numeric features from seven columns (`Period`, `Duration`, `Depth`, `SMass`, `SRad`, `SRadEst`, `Tmag`): three missingness flags; log10 period/duration/depth; duration/period duty cycle; effective stellar radius; log estimated planet radius; raw mass/radius/Tmag. Train-only medians and `StandardScaler` are reused. Approximate radius relation: `Rp/Re ≈ 109.076 × Rstar/Rsun × sqrt(depth_fraction)`; not a measured radius.

Self-supervised encoder (not a classifier)

`src/astrovet/ssl/neural_encoder.py` is NumPy dense network 200→64→16 and decoder 16→64→200. LeakyReLU alpha=.1, He/Kaiming weights, zero biases, L2-normalized inference embedding. It learns from unlabeled views and cannot classify planets by itself.

For target `x`, masked/jittered views `x1,x2`, reconstruction `xhat1`, embeddings `z1,z2`, masked coordinates weighted `m=2` (otherwise 1), objective is `L=mean[m(xhat1-x)^2] + λvmean[(z1-z2)^2]`, `λv=.5`. Mask fraction=.15; jitter sigmas=.02/.04. Manual chain-rule gradients, global norm clip 5, Adam (`β1=.9, β2=.999, ε=1e-8`); CLI defaults 5 epochs, batch 16, lr .005, seed 42. CPU only, no auto-diff, early stopping or scheduler. Views are built by `ssl/dataset.py`; NPZ stores weights/provenance. Loss reduction is not proof of classification quality.

Supervised fusion and signal diagnostics

`scripts/train_ssl_physics_fusion.py`: 16 SSL + 11 physics/missingness features (27; baseline excluded), train-only `StandardScaler`, class-balanced logistic regression C□{.01,.1,1,10}, max_iter=3000, seed42. Validation average precision chooses C; Platt fits disjoint calibration hosts. Labels CP/KP vs FP/FA in follow-up-selected TOIs; not all TESS.

- BLS: scans trial period/phase/duration for box-like dips; Kovács, Zucker & Mazeh (2002). Peaks are not planet identity.
- TLS: scans transit-shaped templates; Hippke & Heller (2019). Sensitivity to small transit shapes does not prevent EB/artifact detections.
- Approximate transit fit: `physics/transit_model.py` uses `scipy.optimize.curve_fit` on approximate limb-darkening template for depth, duration, time offset. Reduced chi-square `Σ[(flux-model)/error]²/(n-3)`. Code explicitly sets `is_validated_physical_inversion=False`; not full Mandel & Agol inversion.
- Odd/even: alternating event depths; Welch test when possible, pooled/normal approximation or abstain. Mismatch may indicate EB; agreement doesn't exclude one.
- Secondary: phase≈.5; project flags SNR≥3, depth ratio≥.08 and >80 ppm. Heuristic, not universal physical law.
- Platt calibration: logistic mapping on logit(raw score) using calibration labels; can't remove selection bias.
- Metrics: ROC-AUC ranking; average precision prevalence-sensitive; Brier `mean((p-y)^2)`; ECE weighted bin gap. Precision=TP/(TP+FP), recall=TP/(TP+FN), F1=2PR/(P+R).

Random Forest, SVM, XGBoost and supervised neural classifiers are not trained in main paths. They are possible comparisons, not current methods. No benchmark proves logistic regression superior.

3. Dataset forensics

Catalog CSV

`artifacts/reports/tces_v3_audit.json` records `tces_with_labels_v3.csv`, 24,926 rows × 23 columns, 3,135,630 bytes, SHA-256 `8c4412b073ac3fb02b8b8ee61be32d25d002cd569d4975f848cf8deddcadc1bb`. Raw CSV is absent; recorded path `D:\Final Year Projet\tces_with_labels_v3.csv` unavailable, so these values are historical, not recomputed today.

Columns: TIC ID, Epoch, Period, Duration, Depth, SMass, SRad, SRadEst, Tmag, Year, MinT, MaxT, Split, Consensus Label, L1-L8, Notes. Recorded label counts: J=15,803; missing=5,340; E=2,613; B=741; S=275; N=154. E/B/J cohort=19,157 rows/19,145 hosts; E prevalence 13.64%; S/N/missing excluded=5,769. Reported missing SMass=12,442 (49.92%), SRad=2,820 (11.31%), SRadEst=579 (2.32%); zero exact duplicates, 38 repeated TIC rows.

`load_and_audit_dataset()` in `models/training.py` checks path, hash/size, reads UTF-8 errors ignored, pandas `read_csv(low_memory=False)` (comments supported), drops recognized units row, validates required columns, coerces TIC, retains E/B/J and maps E→1. Whole CSV loads into pandas memory; no chunk/lazy load. Missing file raises `FileNotFoundError`; bad schema `ValueError`; comma default. E includes eclipsing binaries, not planet truth.

FITS and disposition cohorts

FITS is an astronomy binary container, not necessarily an image. Light-curve FITS stores timestamps, flux, uncertainty, flags and headers; pixel FITS stores spatial arrays. ASTROVET's primary input is the former. Checkout has two 1,958,400-byte copies of a Sector 23 SPOC FITS for TIC 390651552; full QLP archive and fusion bundle are absent.

`docs/implementation_status.md` records prior download: 8,151 TOI rows; 107 MAST TCE catalogs (156,431 rows; 248.09 MB); 18,121 period/epoch matches; hard-label cohort 1,782 events/1,734 hosts (CP 565, KP 489, FP 706, FA 22). One run yielded 1,771 FITS feature rows, 11 missing. Later run removed 20 held-out hosts seen in SSL, producing 1,751 features. D: data/manifests are unavailable; historical figures only. MAST/SPOC/QLP products are public; model/data artifacts are local/ignored by Git.

Limitations: imbalance, missing stellar metadata, label ambiguity/noise, selected TOI follow-up bias, sector/pipeline/cadence shifts. Host-disjoint random split is not temporal or independent-source test.

4. Preprocessing and execution traces

Catalog path

```
24,926 × 23 raw (audit record)
→ E/B/J: 19,157 × 23; 5,769 excluded
→ 19,145 TIC groups; 60/15/10/15 train/validation/calibration/test, seed 42
→ 12 derived features from 7 numeric columns
→ train-only median imputation + StandardScaler
→ fit logistic model on train; choose C by validation PR-AUC
→ fit Platt on calibration; evaluate test once
```

Rows remain constant; columns change. IDs, labels, votes, notes, supplied split and time-boundary fields excluded. Fitted medians/scaler/order stored and reused.

FITS inference

```
Browser upload (/api/upload_and_vet or /api/upload)
→ size/type/signature/readability validation; save original + hash + audit
→ choose SPOC PDCSAP_FLUX or QLP KSPSAP_FLUX
→ finite/quality filtering → gap segmentation/normalization/detrending
→ supplied ephemeris or BLS/TLS → folded transit/full/odd/even views
→ approximate fit, odd/even, secondary, plausibility diagnostics
→ optional SSL embedding + physics vector + optional fitted fusion
→ optional reference catalog or explicit container baseline
→ score/reasons/report/plots; optional TESSCut pixel image
→ JSON to browser; optional Groq prose from sanitized summary
```

Prior cleaning test injected a 1000-ppm box into each of two real files and recovered depth for those cases. This does not prove general transit preservation over periods/noise/gaps/cadences. Original upload is retained unchanged.

Workflow map (Mermaid source)

```
flowchart TD
 A[python -m astrovet.cli ui] --> B[Click ui_cmd / run_dashboard_server]
 C[Browser FITS upload] --> D[do_POST validate/store/audit]
 D --> E[_execute_vetting]
 E --> F[vet_and_generate_report]
 F --> G[read/filter/normalize]
 G --> H[BLS TLS + folding]
 H --> I[physics diagnostics]
 I --> J[optional SSL/fusion]
 I --> K[optional ExoMiner]
 J --> L[score/reasons/reports/plots]
 K --> L
 L --> M[optional Groq prose]
 M --> N[JSON to browser]
```

Function contracts

Function	Input → operations → output
`cli()`/`ui_cmd()` in `src/astrovet/cli.py`	host/port → local HTTP server; console entry `astrovet=astrovet.cli:main`.
`do_POST()` in `ui/server.py`	upload + metadata → validate/store/audit → `_execute_vetting`; arbitrary paths rejected.
`_execute_vetting()`	candidate/FITS/optional ephemeris → report pipeline → JSON and artifact links.
`read_tess_fits_lightcurve()`	FITS → time/flux/error/quality/header data.
`filter_quality_cadences()` / `segment_and_normalize_lightcurve()`	arrays → retained cadences, segments and normalized curve/audit.
period search in `views/periodogram.py`	series + optional ephemeris → BLS/TLS outputs/agreement.
`run_physics_diagnostics()`	folded/raw series → fit and consistency checks.
`SelfSupervisedTransitEncoder.encode()`	length-200 view(s) → median/MAD cleaning → 16-D vector.
`AstrovetFusionModel.predict_single()`	optional baseline + physics + embedding → fitted score or fallback triage.

Function	Input → operations → output
`generate_groq_explanation()`	compact JSON + env key → structured prose, non-fatal.
`run_training_pipeline()`	CSV → audit/splits/transforms/tuning/calibration/test/report/joblib.
`scripts/train_ssl_physics_fusion.py:main()`	cohort/manifest/split/checkpoint → features, model, calibration, metrics.

5. Results and their limits (not rerun)

Catalog E-vs-B/J saved report

Recorded host counts train/validation/calibration/test: 11,487/2,871/1,914/2,873; events: 11,489/2,874/1,916/2,878; seed 42 and zero overlap. Selected C=10. Test N=2,878: 390 E, 2,488 B/J, prevalence .1355. ROC-AUC .8901; average precision .5368; raw→calibrated Brier .1417→.0825; ECE .2027→.0222. At threshold .5: precision .6069, recall .4077, F1 .4877; confusion `[[TN 2385,FP 103],[FN 231,TP 159]]`. Precision@10=.50, @25=.60, @50=.68; recall-at-precision ≥80% entries zero. These are catalog label metrics, not planet accuracy.

Implied accuracy 88.4%; majority-negative baseline 86.45%, so accuracy alone is weak. Long-period PR-AUC recorded .1296 (15 positive of 969); shallow <500 ppm .1782. No CIs, repeated seeds or independent temporal test reported.

Full QLP SSL+fusion report

Historical docs report 11,511 FITS/11,487 train hosts; 1,171,623 views; five CPU epochs, batch 16; loss .765910→.760074. Fusion cohort 1,751 rows; partitions train 1,061 events/1,033 hosts, validation 266/256, calibration 168/163, test 256/251. Test ROC-AUC .8039, PR-AUC .8377 at prevalence .5781; Brier .1819→.1795; ECE .0983→.0825; threshold .5 precision .7322, recall .9054, F1 .8097; confusion `[[59,49],[14,134]]`. Selected follow-up cohort only.

Artifact distinction: Full D: model bundle is unavailable here. Checkout has three small real-FITS NPZ checkpoints (single-file provenance says 135 views) and catalog joblib. It lacks the documented matching full fusion model. These metrics do not prove today's UI loads it.

Leakage and deployment

Controls: TIC-grouped partitions; IDs/labels/votes/notes/time boundaries excluded; train-only transforms; validation C selection; calibration-only Platt; untouched test; SSL train-host manifest option; later report documents held-out host removal; checkpoint compatibility checks; missing baseline not fabricated.

Risks: random group split is not future sector/source test; E conflates planets/binaries; TOIs are selected; no CIs/repeated seeds; narrow injections; artifacts may be absent; one historical FITS shorter than header was parsed. No full security audit. Inference needs compatible local artifacts; otherwise heuristic/unavailable. Groq is called after local result, gets coarse non-identifying fields only, cannot change score. Render prose as text and keep API key environment-only.

Pending verification checklist

1. Restore/version exact full encoder, fusion joblib, cohort/manifests; verify hashes, compatibility and dashboard launcher.
2. Rerun full tests after recent changes (not run in this audit).
3. Compare browser/CLI/report on broader labeled SPOC/QLP; assert score/model/calibration provenance parity.
4. Run injection/recovery grid over depth, period, duration, variability, noise, gaps, flags, cadence, sector, pipeline.
5. Freeze model/threshold then evaluate blinded independent/temporal/pipeline cohort; report PR, precision/recall at review capacity, calibration, CIs, subgroups and coverage.
6. Revalidate calibration at deployment prevalence; no population-wide probability claim.
7. If live ExoMiner required, demonstrate complete real inference within memory; else report unavailable.
8. Verify Groq with rotated key and inspect transmitted data; never commit key.

6. Annotated dataset-loading code

From `load_and_audit_dataset()` in `src/astrovet/models/training.py`:

```
p = Path(file_path)
if not p.is_file():
    raise FileNotFoundError(...)
file_size = p.stat().st_size
file_sha256 = compute_sha256(p)
with open(p, "r", encoding="utf-8", errors="ignore") as f:
    first_line = f.readline().strip()
if first_line.startswith("#"):
```

```
    df_raw = pd.read_csv(p, comment="#", low_memory=False)
else:
    df_raw = pd.read_csv(p, low_memory=False)
...
df_raw["TIC ID"] = pd.to_numeric(df_raw["TIC ID"], errors="coerce").astype(np.int64)
valid_mask = df_raw["Consensus Label"].isin(["E", "B", "J"])
df_valid = df_raw[valid_mask].copy().reset_index(drop=True)
df_valid["target"] = (df_valid["Consensus Label"] == "E").astype(int)
```

It verifies file/provenance, loads the whole CSV in memory, supports comments/units row, validates schema, converts TIC, excludes uncertain labels and creates E-vs-B/J target. FITS reading is separate.

7. Project-specific viva question bank (100)

Answers include a simple answer then a technical anchor. File paths are relative to the repository.

A. Basic project questions (15)

1. What is ASTROVET? Simple: TESS candidate triage. Technical: FITS→search/diagnostics→optional SSL/fusion→report; `ui/server.py:_execute_vetting`, `explain/report.py`. Verified.
2. What problem? Transit-like dips have non-planet causes; multiple clues aid human review. `physics/*.py`.
3. Who uses it? Researchers/students/vetters; UI is human-review oriented. No user study.
4. Why important? Many events need review; time saved is not measured.
5. What if unsolved? Manual vetting continues; impact not quantified.
6. Existing approach? Search pipelines plus human/ML vetting; ASTROVET integrates searches and adapter. Verified high-level.
7. Existing disadvantages? Model/label scope can limit use; do not claim all other methods fail. `model_card.md`.
8. Proposed approach? Physics evidence plus SSL and supervised fusion. Encoder/fusion files.
9. What is different? Integrated auditable workflow; superiority unproven.
10. Contribution? Implemented prototype, not a new BLS/TLS method.
11. New or improvement? New project workflow built on established methods; ExoMiner external.
12. Input/output? FITS + optional ephemeris → evidence/triage/report. Server/schemas.
13. Assumptions? FITS metadata usable and calibration data matches deployment; latter is risky.
14. Boundaries? Screening only; no follow-up or confirmation.
15. Done/partial? Core code exists; broad validation/artifact availability open.

B. Machine learning (20)

16. Supervised catalog algorithm? Logistic regression; `models/training.py`.
17. Target? E vs B/J, not planet; model card.
18. Equation? `σ(w■x+b)`.
19. Loss? Class-weighted regularized binary log loss; sklearn implementation.
20. C? Inverse regularization; saved winner 10.
21. Optimizer? `lbfgs`, max_iter 2000.
22. Why logistic? Efficient, simple fixed-width baseline; not proved best.
23. Why not Random Forest? Not trained; nonlinear baseline remains to compare.
24. Why not SVM? Not trained; probability calibration/scale need comparison.
25. Why not XGBoost? Not in main training path; no benchmark.
26. Neural classifier? Encoder is neural, fusion head logistic; do not conflate.
27. SSL means? self-generated reconstruction/view targets without labels.
28. What learned? MLP/decoder weights, not planet classes.
29. SSL loss? reconstruction MSE + .5 consistency MSE, mask weight 2.
30. Optimizer? manual backprop, clip 5, Adam .9/.999/1e-8, lr .005.
31. Parameters/hyperparameters? weights learned; C/lr/epochs/dimensions configured.

32. Fusion? 16 SSL + 11 physics inputs, standardized logistic head.
33. Platt? logistic mapping fitted on disjoint calibration set.
34. Compute? CPU NumPy/sklearn; full performance benchmark absent.
35. 10× data? batching exists, but view extraction may use memory; streaming needed.

C. Dataset questions (20)

36. What is CSV? Astronet-Triage-v2 TCE catalog; audit reports 24,926×23; raw absent.
37. Origin? Project cites Tey 2023/Zenodo; exact file unavailable now.
38. Columns? TIC, transit/star values, labels/votes/time/notes; audit JSON.
39. Classes? E/B/J/S/N/null; only E/B/J retained.
40. Why E not planet? Includes binaries; code and paper.
41. Why exclude S/N/null? uncertain/unlabeled, not safe negatives.
42. Balanced? No; E prevalence 13.64% in usable rows.
43. Missingness? SMass ~49.92%; flags+train medians.
44. Duplicates? Audit says zero exact rows, 38 repeated TIC; not rechecked raw.
45. SSL data? real FITS; full QLP run documented, not present.
46. Fusion cohort? CP/KP vs FP/FA selected TOIs.
47. Source? MAST TESS; downloads remain local.
48. Why FITS? precise time-series flux/quality needed for transit search.
49. Pixel image input? no, supplemental context only.
50. Loaders? pandas CSV and FITS reader separate.
51. Batch CSV? no, whole DataFrame.
52. Encoding/delimiter? UTF-8 errors ignored, comma default, optional comments.
53. Missing file? FileNotFoundError; schema ValueError.
54. Public? sources public, artifacts not all tracked.
55. Bias/noise? plausible/documented; no quantified label-noise study.

D. Preprocessing/features (15)

56. CSV transforms? numeric coercion, logs, duty cycle/radius features.
57. Missing handling? flags plus train medians.
58. Why scaling? disparate units; train-only to avoid leakage.
59. Dimension change? seven columns→12 features, same rows.
60. Flux fields? docs specify SPOC PDCSAP/QLP KSPSAP; verify fallback before viva.
61. Noise filtering? finite/quality filtering, segmentation/detrending; not all artifacts.
62. Could cleaning erase transit? yes; two isolated injections only, broader grid pending.
63. Why segment gaps? avoid trends across discontinuities.
64. SSL scaling? median/MAD, interpolation, clip ±8.
65. Dimensions? 200 bins, embedding 16.
66. Split timing? before fitting transforms; fusion scaler train-only.
67. Inference reuse? stored scaler/medians/order and checkpoint.
68. Leakage? train-only and host manifest safeguards; verify each run.
69. Schema drift? validate/fail, don't fabricate.
70. Remaining test? multi-factor injections across pipeline/cadence/gaps.

E. Training and evaluation (15)

71. Catalog entry? CLI `model train`→`run_training_pipeline()`.
72. SSL entry? `ssl pretrain`→`ssl_pretrain_cmd`→encoder.fit.
73. Fusion entry? `scripts/train_ssl_physics_fusion.py:main()`.
74. Epoch/batch? pass/update group; 5 epochs, batch16 defaults.
75. Early stopping? none in SSL.
76. Checkpoints? NPZ encoder, joblib classifier.
77. Selection? validation PR-AUC chooses C; test untouched.
78. Test set? 15% host split recorded; not necessarily future data.
79. Metrics? PR/ROC, P/R/F1, Brier/ECE, subgroups, precision@k.
80. Confusion? 2385 TN,103 FP,231 FN,159 TP at .5.
81. Accuracy? 88.4% vs 86.45% majority; insufficient alone.
82. PR-AUC? .5368 vs .1355 prevalence; not planet accuracy.
83. Confidence intervals? not reported; group bootstrap recommended.
84. Repeated runs? not reported; seed42 single run.
85. Reproducible? hashes/manifests help, but raw/full model absent from Git.

F. Architecture (10)

86. First code? `astrovet.cli:main` console entry.
87. Upload flow? `do_POST`→audit→`_execute_vetting`→report→JSON.
88. Model load scope? don't assert once/request without verifying report path.
89. Baseline absent? unavailable, not fabricated zero.
90. Output path? `artifacts/reports`, reports/PNGs.
91. Docker purpose? isolate upstream; not GPU acceleration; OOM recorded.
92. Groq? optional prose with environment key; live response unverified.
93. Does Groq change score? no.
94. Upload security? validation/path checks, not full security review.
95. Bad FITS? validation error; synthetic upload fallback disabled.

G. Research questions (5)

96. Research gap? integration is motivation; formal literature review absent.
97. Evidence? established methods, selected cohort metrics, two real-file checks; broader test absent.
98. Domain shift? warnings/availability flags exist; drift adaptation unvalidated.
99. Prove value? frozen blind benchmark and reviewer-capacity/injection metrics; pending.
100. More time? restore artifacts, blind cohort, injection grid, CIs, cross-pipeline.

8. Twenty challenging follow-ups

1. E contains binaries, why planet classifier? It isn't; separate catalog signal label task.
2. ROC .89 but recall .41? ROC ranks thresholds; report operating point separately.
3. Calibration generalizes? Unknown outside cohort; test at deployment prevalence.
4. Could test hosts occur in SSL? yes unless manifest enforced; check run audit.
5. SSL labels? no; fusion labels yes.
6. Why SSL objective? proxy reconstruction/invariance; downstream test still required.
7. Does loss prove quality? no.

8. Clip ±8 physical? no, numerical guard; sensitivity study needed.
9. Preserve long transits? not established; grid pending.
10. BLS/TLS disagree? route for review, retain disagreement.
11. No secondary rules out EB? no.
12. Why 0.08 secondary ratio? heuristic, validate empirically.
13. Physical transit fit? approximate, explicitly not inversion.
14. Why unlabeled vs labeled counts differ? SSL and supervised roles differ.
15. Independent test? host-disjoint, not necessarily time/source independent.
16. TIC duplicates leak? grouping keeps host together.
17. Why omit TIC feature? avoid identifier memorization.
18. Schema changes? should error, not guess.
19. Structured Groq output truth? only format guaranteed, prose untrusted.
20. Complete? implementation is substantial; broad validation/artifact availability open.

9. Mock viva

ML researcher

Q: Why logistic and are probabilities reliable? Answer: “Catalog baseline uses class-weighted logistic regression over 12 features; validation PR-AUC chooses C and separate hosts fit Platt. Target is E-vs-B/J. Recorded PR-AUC .5368 at .1355 prevalence and calibrated Brier .0825, only on this cohort. Need independent validation and intervals.” Evidence: `models/training.py`, `calibration.py`, `docs/model_card.md`.

Q: Explain SSL. Answer: “NumPy 200-64-16 encoder/decoder reconstructs masked views and aligns jittered embeddings without labels; downstream fusion uses labels. Loss drop alone is not class quality.” Evidence: `ssl/neural_encoder.py`.

Q: Leakage? Answer: “TIC grouping, train-only transforms, validation tuning, calibration-only Platt and untouched test. Newer run documents held-out hosts removed from SSL. Check exact manifest; this is not future-sector testing.”

Study: label scope, thresholds, PR prevalence, calibration, host overlap, confidence intervals.

Software engineer

Q: Trace upload. Answer: “Click starts HTTP server; `do_POST` validates/stores/audits; `_execute_vetting` calls report pipeline and serializes JSON. Invalid data errors; optional missing output remains unavailable.” Evidence: `cli.py`, `ui/server.py`, validator.

Q: Model pairing? Answer: “Bundle records feature names/scaler and compatibility guard exists, but full bundle is absent from this checkout, so runtime pairing cannot be proven.” Evidence: `models/fusion.py` and inventory.

Q: Groq? Answer: “Only coarse wording summary leaves; environment key; prose can't score. Live request with rotated key still needs verification.” Evidence: `groq_explainer.py`.

Study: load scope, failures, upload limits, artifact versioning.

External examiner

Q: Novelty? Answer: “Integrated established searches/diagnostics/SSL into auditable triage, not new BLS/TLS or proven ExoMiner improvement. Current evidence is selected cohort and narrow smoke cases.”

Q: General TESS? Answer: “No; follow-up cohort selection shifts prevalence. Treat as review priority until independent test.”

Q: Main limitation? Answer: “External validation and artifact reproducibility; need blind cross-pipeline data, injection grid and full model bundle.”

Avoid: confirmed planet claims; conflating CSV and FITS models; ROC as calibration; Groq as detector; no-secondary as proof; TESS-wide claims from two files.

10. Sources and core answers

Repository sources

`README.md`; `docs/implementation_status.md`, `model_card.md`, `validation_results.md`, `data_dictionary.md`, `label_policy.md`, `upload_workflow.md`; `src/astrovet/cli.py`; `models/training.py`, `fusion.py`, `calibration.py`; `ssl/neural_encoder.py`, `dataset.py`; `lightcurves/reader.py`, `quality.py`, `normalization.py`, `validator.py`; `views/periodogram.py`, `folding.py`; `physics/*.py`; `baseline/adapter.py`, `container.py`; `explain/report.py`, `pixel_preview.py`, `groq_explainer.py`; scripts for cohort and training.

Authoritative external references

- MAST TESS mission archive
- NASA/HEASARC TESS data products
- NASA/HEASARC TESS documentation/SPOC
- MIT TESS QLP
- Tey et al. 2023 Astronet-Triage-v2, Zenodo 7411579
- NASA ExoMiner, ExoMiner++ paper
- Astropy BLS documentation
- Hippke & Heller 2019 TLS
- Groq chat, structured outputs, API reference

Final answers to memorize

1. Built: TESS candidate triage with FITS processing, BLS/TLS, physics checks, SSL features, optional supervised fusion, human-readable evidence.
2. Purpose: organize human review of transit-like signals with multiple causes.
3. Flow: upload→filter/normalize→search/physics→optional SSL/fusion/baseline→triage/report.
4. ML: NumPy SSL encoder; separate sklearn logistic catalog and fusion heads; Platt calibration.
5. Data: MAST FITS, Astronet-Triage-v2 catalog, matched TOIs; local artifacts aren't all in Git.
6. Evaluation: host-held-out test but selected cohort; no broad independent validation.
7. Limitations: ambiguous target, follow-up selection, missing full model in checkout, narrow injections.
8. Next: restore artifacts, rerun tests, blind validation and injection grid.
9. Groq: optional wording only; no score effect.

Verification statement: This is a code/artifact audit, not a rerun of every training job. Historical metrics are explicitly prior records. Raw CSV and full fusion model were unavailable. Current full test suite and live Groq request were not run.